

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2025-2026
Mid-Semester Test
(EC-2 Regular/ SET 2)

Course No. : AIMLCZG511
Course Title : DEEP NEURALNETWORKS
Nature of Exam : Closed Book
Weightage : 30%
Duration : 2 Hours
Date of Exam : 21st JUNE 2026(AN)

No. of Pages =
No. of Questions = 6

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q1. A media technology company is independently developing two AI-based predictive models to analyze and anticipate subscriber behaviour across OTT (Over-The-Top) streaming platforms:

Model A — Churn Prediction Model: The first model is designed to predict whether an existing subscriber of SonyLive OTT is likely to deactivate(churn) their account or not.

Model B — Platform Preference Prediction Model: The second model aims to predict which OTT platform a user is most likely to subscribe to, given a choice among three competing platforms — Oho, Sunny and Joeo.

[5 Marks]

- a). Explain why Deep Learning would be preferred over traditional Machine Learning algorithms for predicting subscriber churn (Model A) and platform preference (Model B) when dealing with large-scale OTT subscriber datasets. What advantages does it offer?
- b). For Model A (Churn Prediction), design a basic Artificial Neural Network (ANN) architecture by specifying: Number of inputs, hidden, and output layers, recommended activation functions for each layer. Justify your design choices.
- c). For Model B (Platform Preference — Oho, Sunny, Joeo), which output layer configuration would you choose and why?
 - i) How many neurons should the output layer contain?
 - ii) Which activation function is most appropriate — Sigmoid, Softmax, or ReLU? Justify.

Q2. Design and visualize a Deep Feedforward Neural Network (DFNN) based on the architectural specifications provided below. For each layer in the network, clearly illustrate the connections and compute the total number of trainable parameters.

[5 Marks]

Layer	Nodes / Neurons	Activation Function
Input Layer	4	-
Hidden Layer 1	6	ReLU
Hidden Layer 2	5	ReLU
Hidden Layer 3	4	ReLU
Output Layer	3	Softmax

Why is Softmax used at the output layer instead of Sigmoid? Under what condition would Sigmoid be more appropriate at the output layer?

Q3. Consider the following Python code snippet to train a deep neural network on the MNIST handwritten digit dataset using Keras:

[5 Marks]

```
from tensorflow.keras.datasets import mnist
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten
from tensorflow.keras.utils import to_categorical

# Load data
(x_train, y_train), (x_test, y_test) = mnist.load_data()
x_train, x_test = x_train / 255.0, x_test / 255.0
y_train, y_test = to_categorical(y_train, 10), to_categorical(y_test, 10)

# Build model
model = Sequential([
    Flatten(input_shape=(28, 28)),
    Dense(128, activation='relu'),
    Dense(64, activation='relu'),
    Dense(10, activation='softmax')
])

model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
model.fit(x_train, y_train, epochs=5, batch_size=32, validation_split=0.1)
```

- a). First layer of the model is: Flatten (input_shape=(28, 28))
- i) What is the role of the Flatten layer in this network?
- ii) What is the output shape of the Flatten layer?
- b) Why does the output layer have exactly 10 neurons?
- c) Why is Softmax the appropriate activation function for this layer?
- d) What would the output of this layer look like for a single input image? Give an example.
- e) Calculate the total number of trainable parameters in the network layer by layer

Q4. A data scientist is building a fraud detection system to classify financial transactions as either **Fraudulent** or **Legitimate**. When the transaction features are plotted in 2D space, it is observed that:

- The two classes form overlapping, concentric circular clusters
- No single linear boundary can separate them effectively

Answer the following:

[5 Marks]

- a) Why would a Single-Layer Perceptron completely fail in this fraud detection scenario? What is the mathematical reason behind this limitation?
- b) Propose an MLP architecture suitable for this problem. Justify your choice of:
- Number of hidden layers and neurons
 - Activation functions at hidden and output layers
 - How the network learns non-linear boundaries through backpropagation
- c) If the fraud detection model needs to classify transactions into 4 categories (Legitimate, Minor Fraud, Major Fraud, Identity Theft), how would the output layer of the perceptron network change? Calculate the minimum number of perceptrons needed across the network.

Q.5. You are developing an Image Recognition Classifier capable of identifying 5 object categories: Cat, Dog, Car, Bird, and Horse. For a given input image, your neural network produces the following raw logit scores at the output layer:

$$Z = [2.5, 0.3, -1.2, 3.1, 0.7]$$

Class Order: [Cat, Dog, Car, Bird, Horse]

[5 Marks]

- Using the Softmax activation function, compute the predicted probability distribution $\hat{y}$ for each class. Round your answers to two decimal places.
- The true class label for this image is "Bird" (4th class). Compute the Cross-Entropy Loss.
- Design a Fully Connected Neural Network (FCNN) for this 5-class image recognition problem with the following specifications and also compute the total number of trainable weights in this network.

Input Features: 20

Hidden Layer 1: 15 neurons

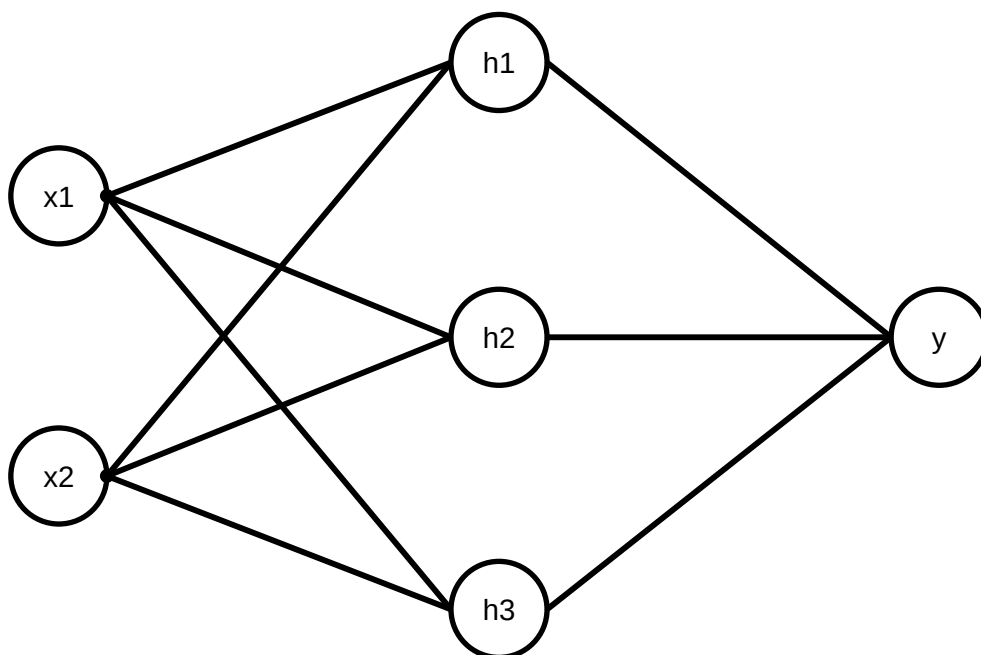
Hidden Layer 2: 10 neurons

Output Layer: 5 neurons

Biases excluded

Q.6. Consider the following DFNN Architecture.

[5 Marks]



Inputs: $x_1 = 0.5$, $x_2 = 0.3$ and output: $y = 1$

Hidden layer weights: $w_{11}=0.4$, $w_{12}=0.3$, $w_{13}=0.2$, $w_{21}=0.2$, $w_{22}=0.6$, $w_{23}=0.4$

biases: $b_1=0.1$, $b_2=0.2$, $b_3=0.1$

Output layer weights: $w_{31}=0.5$, $w_{32}=0.7$, $w_{33}=0.7$ | bias: $b_4=0.3$

Activation: Sigmoid at each neuron.

- Compute the predicted value ($\hat{y}$) of this architecture
- Calculate Loss by considering MSE as Loss Function

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